

Unsupervised Spectroscopic Chemometrics Handout: Decoupling High-Dimensional Data Matrices via PCA and t-SNE

Llinersy Uranga Piña, LPPI - Cergy-Paris Université

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1 Introduction: The Dimensionality Curse in Spectroscopy

Modern analytical spectroscopy methods including Attenuated Total Reflection Fourier-Transform Infrared (ATR-FTIR), Raman, and X-ray Diffraction (XRD) yield complex, high-dimensional datasets. A single scanning run can record hundreds or thousands of independent absorbance steps (wavenumbers) per sample. Manually comparing these files using standard single-peak height ratios introduces observer bias and discards valuable multi-component spectral information.

The computational architecture implemented in **Module 3** addresses this complexity through unsupervised machine learning frameworks. This handout explains the physical, mathematical, and chemometric mechanisms behind **Principal Component Analysis (PCA)**, **Loading Vectors**, and **t-SNE manifold projections**, demonstrating how to extract clear material properties from high-dimensional spectral noise.

2 Matrix Standardization & Preprocessing

Before applying dimensionality reduction algorithms, raw spectroscopic data matrices must undergo rigorous preprocessing to ensure that physical variances drive the models rather than instrumental artifacts.

2.1 1. Matrix Realignment & Transposition

Raw data exported from spectrometers typically tracks intensity against variable wavenumbers. To align multiple data documents into an un-biased, uniform feature space, the pipeline transposes the data grid:

```
1 X_raw = spectral_matrix.T
```

This changes the core matrix geometry from a wavenumber-dominant orientation to a standard observation matrix configuration with dimensions of $N \times P$, where N represents the independent sample runs and P represents the uniform feature coordinates.

2.2 2. Z-Score Mean Standardization

Variations in crystal contact, film thickness, or source focus drift can introduce broad vertical scaling offsets across sample spectrum traces. The code corrects these artifacts using a standard Z-score transformation layer via **StandardScaler**:

$$Z_{ik} = \frac{X_{ik} - \mu_k}{\sigma_k} \quad (1)$$

where μ_k is the mean intensity of wavenumber channel k and σ_k is its standard deviation. This shifts the model’s focus from absolute intensity offsets to localized peak changes, preventing highly prominent bands from dominating the variance mapping.

3 Principal Component Analysis (PCA)

PCA compresses the standardized data matrix by rotating the original coordinate frame onto a new system of orthogonal axes known as Principal Components (PCs).

3.1 1. Covariance Formulations

The pipeline evaluates the correlations between all wavenumber channels by constructing a $P \times P$ covariance matrix $\mathbf{C}$:

$$\mathbf{C} = \frac{1}{N-1} \mathbf{Z}^T \mathbf{Z} \quad (2)$$

To locate the principal axes of variance, the system computes the characteristic eigenvector decomposition:

$$\mathbf{C} \mathbf{v}_i = \lambda_i \mathbf{v}_i \quad (3)$$

where λ_i represents the eigenvalue quantifying variance along the directional eigenvector coordinate axis $\mathbf{v}_i$.

3.2 2. The Variance Threshold Rule (Lab Task 1 Insight)

A common mistake among experimentalists is assuming that the first two principal components (PC1 and PC2) are always sufficient for complete sample characterization. While excellent for rendering a flat 2D projection map, printing the cumulative explained variance reveals their limitations:

```
1 var_exp = np.cumsum(pca.explained_variance_ratio_)
```

In complex datasets containing subtle chemical structural variations and high baseline noise, the first two PCs may capture only a small fraction of the variance (e.g., 26.29%). To safely clear a research-grade **90% variance threshold**, an experimentalist must incorporate additional higher-order components (PC1 $\rightarrow$ PC4), demonstrating that subtle chemical properties often hide in secondary components.

4 Physical Interpretation of Eigenvectors (Loadings)

The absolute core power of PCA for experimental researchers lies in the analysis of the **Loading Coefficients**, which map the importance of each original wavenumber back onto the principal components.

4.1 1. Wavenumber Importance Tracking (Lab Task 2 Insight)

By plotting loading coefficients against the wavenumber spectrum, researchers can determine the exact chemical properties driving group separations:

```
1 plt.plot(wavenumbers, pca.components_[0], label='PC1 Loadings')
```

In an engineered or contaminated dataset:

- **PC1 Loadings** consistently show a broad peak spanning the 3200–3600 cm^{-1} envelope. This matches the broad water or hydroxyl ($-\text{OH}$) stretching region, showing that PC1 separates synthesis failures based on broad impurity additions.

- **PC2 Loadings** exhibit a sharp peak at 1710 cm^{-1} . This maps perfectly to the carbonyl (C = O) stretching vibration, proving that PC2 tracks photo-oxidative degradation trends across the sample set.

This extraction transforms PCA from a blind clustering algorithm into an analytical tool that directly identifies the chemical functional groups responsible for sample variance.

5 Unsupervised Clustering: K-Means Over-Segmentation

5.1 The Clustered Constraint Optimization

When analyzing unknown samples, the K-Means algorithm partitions data into K groups by minimizing the within-cluster sum-of-squares distance from the centroids μ_j :

$$\arg \min_{\mathbf{S}} \sum_{j=1}^K \sum_{\mathbf{x} \in S_j} \|\mathbf{x} - \mu_j\|^2 \quad (4)$$

5.2 The Risk of Arbitrary Partitioning (Lab Task 3 Insight)

Setting an unphysical cluster count (e.g., forcing $K = 5$ on a system containing only 3 real chemical phases) causes severe over-segmentation. The algorithm cuts unified, contiguous sample groups into arbitrary mathematical slices that have no physical or chemical meaning.

To confirm real physical sample phases, experimentalists should run an Elbow Plot (tracking Inertia vs. K) or evaluate Silhouette Coefficients to ensure cluster counts reflect true chemical variations rather than mathematical artifacts.

6 Non-Linear Manifold Projections: t-SNE vs. PCA

6.1 1. The Global vs. Local Geometry Paradigm (Lab Task 4 Insight)

While PCA applies linear rotations to maximize global variance across the entire coordinate space, t-Distributed Stochastic Neighbor Embedding (t-SNE) handles non-linear relationships by mapping similarities into tight, isolated clusters.

6.2 2. Probability Workspace Formulations

In high-dimensional spaces, t-SNE converts Euclidean distances between samples into conditional probabilities $p_{j|i}$ that represent similarity:

$$p_{j|i} = \frac{\exp(-\|\mathbf{x}_i - \mathbf{x}_j\|^2 / 2\sigma_i^2)}{\sum_{k \neq i} \exp(-\|\mathbf{x}_i - \mathbf{x}_k\|^2 / 2\sigma_i^2)} \quad (5)$$

The low-dimensional 2D map uses a heavy-tailed Student-t distribution (q_{ij}) to alleviate crowding issues. The final coordinates are optimized by minimizing the Kullback-Leibler (KL) divergence between these probability spaces:

$$\text{KL}(\mathbf{P}||\mathbf{Q}) = \sum_i \sum_j p_{ij} \log_2 \left(\frac{p_{ij}}{q_{ij}} \right) \quad (6)$$

6.3 3. Research Application of Topologies

- **PCA Strengths:** Maps continuous processes well. It displays systematic degradation vectors across a coordinate plane, showing how pristine samples gradually transition into degraded states.
- **t-SNE Strengths:** Groups discrete regional clusters effectively. It pulls localized datasets into highly separated "islands," making it an exceptional tool for clear-cut classification pipelines, such as verifying the geographic origins of natural products.